Jacob M. Chen

jchen459@jhu.edu GitHub: jacobmchen https://jacobmchen.github.io

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Education

- **Johns Hopkins University**, Baltimore, MD — July 2024 – Present
  M.S. and Ph.D. in Computer Science; Advised by Dr. Ilya Shpitser.
- **Williams College**, Williamstown, MA — Sept. 2019 – Dec. 2023
  B.A. Major in Computer Science & Concentration in French and Francophone Studies
  - Thesis in Computer Science: *Causal Inference with Outcome-Dependent Missingness And Self-Censoring*; Advised by Dr. Rohit Bhattacharya
  - *Magna Cum Laude* and *Phi Beta Kappa*

Publications

1. **Jacob M. Chen**, Michael Oberst, "Just Trial Once: Ongoing Causal Validation of Machine Learning Models," in Proceedings of the 41st Conference on Uncertainty in Artificial Intelligence (UAI), 2025.
   - Acceptance rate 230/750=30.7%. Chosen for oral presentation 20/230=8.7%.
   - Link to manuscript: `https://proceedings.mlr.press/v286/chen25b.html`
   - Accompanying code: `https://github.com/jacobmchen/just_trial_once`
2. Jonathan Zhang, Erik Skalnes, **Jacob M. Chen**, Michael Oberst, "Bounding the Causal Impact of ML-assisted Decision-Making via Counterfactual Correctness," in Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI), 2026.
   - Link to manuscript: `https://arxiv.org/abs/2607.21806`
3. Ellen M. Mowry, Peiqing Qian, William Meador, Sharon Lynch, Ram Narayan, Aimee Borazanci, Derrick Robertson, Sara Qureshi, Jennifer Graves, Jason Silversteen, Megan Esch, Leticia Tornes, Claire Riley, Marwa Kaisey, Nancy Sicotte, Siddharama Pawate, Tirisham Gyang, Torge Rempe, Ahmed Z. Obeidat, Geeta Ganesh, Lana Zhovtis Ryerson, Tyler Smith, John Rose, Aaron Carass, Ornusa Chalayon, Mason Kruse-Hoyer, Monilola Salami, Amy Liu, Blake Dewey, **Jacob M. Chen**, Elizabeth Ogburn, Carolyn Koch, Jerry Prince, Sandra D. Cassard, Scott D. Newsome, "The TRaditional versus Early Aggressive Therapy for MS (TREAT-MS) trial: Design and baseline characteristics of participants," in Contemporary Clinical Trials, 2025.
   - Link to manuscript: `https://www.sciencedirect.com/science/article/abs/pii/S1551714425003118`
4. **Jacob M. Chen**, Rohit Bhattacharya, Katherine A. Keith, "Proximal Causal Inference with Text Data," in Advances in Neural Information Processing Systems 38 (NeurIPS 2024).
   - Link to manuscript: `https://proceedings.neurips.cc/paper_files/paper/2024/hash/f53a37f820d5be5930415d964f4a0187-Abstract-Conference.html`
   - Accompanying code: `https://github.com/jacobmchen/proximal_w_text`
5. Julie Joss, **Jacob M. Chen**, Nicolas Prost, Gaël Varoquaux, Erwan Scornet, "On the consistency of supervised learning with missing values," in Statistical Papers, 2024.
   - Link to manuscript: `https://link.springer.com/article/10.1007/s00362-024-01550-4`
   - Accompanying code: `https://github.com/jacobmchen/supervised_missing`
6. **Jacob M. Chen**, Daniel Malinsky, Rohit Bhattacharya, "Causal Inference with Outcome-Dependent Missingness and Self-Censoring," in Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence (UAI), 2023
   - Acceptance rate 243/778=31.2%. Chosen for a poster spotlight presentation (32/243=13.2%).
   - Link to manuscript: `https://proceedings.mlr.press/v216/chen23f.html`
   - Accompanying code: `https://github.com/jacobmchen/mnar-recoverability`

Research Interests

- Causal Inference, Missing Data, Graphical Models, Marginal Structural Models, Applications in Healthcare and the Social Sciences, Computer Science Education

Presentations & Talks

1. **Jacob M. Chen**, Ilya Shpitser. "From Efficient Counterfactual Mean Estimation to Estimating Marginal Structural Models." Poster presentation at the American Causal Inference Conference (ACIC), Salt Lake City, UT, May 2026.
   – Link to poster: `https://jacobmchen.github.io/research/poster_acic2026.pdf`
2. **Jacob M. Chen**, Shivalika Khanduja, Zachary Darby, Joseph DiNatale, Marc Sussman, Diane Alejo, Steven Menez, Lee Goeddel, Ilya Shpitser, and Glenn Whitman. "Acute Kidney Injury During X-Clamp in Cardiopulmonary Bypass Surgery: A Novel Causal Machine Learning Framework for Analyzing Causes, Mechanisms, and Prevention." Poster presentation at Society for Thoracic Surgeons (STS), New Orleans, Louisiana, Jan. 28 - Feb. 1, 2026.
   – Link to poster: `https://jacobmchen.github.io/research/poster_sts2026.pdf`
3. **Jacob M. Chen** and Michael Oberst. "Just Trial Once: Ongoing Causal Validation of Machine Learning Models." Oral presentation at Uncertainty in Artificial Intelligence (UAI), Rio de Janeiro, Brazil, July 21 - July 25, 2025.
4. **Jacob M. Chen**, Rohit Bhattacharya, and Katherine A. Keith. "Proximal Causal Inference with Text Data." Poster presentation at Advances in Neural Information Processing Systems 38 (NeurIPS 2024), Vancouver, Canada, December 2024.
5. **Jacob M. Chen** and Michael Oberst. "Generalizable Randomized Trial Designs for Machine Learning Tools in Healthcare." Poster presentation at the 2024 Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, MD, December 2024.
6. **Jacob M. Chen**, Rohit Bhattacharya, and Katherine A. Keith. "Proximal Causal Inference with Text Data." Poster presentation at the American Causal Inference Conference (ACIC), Seattle, WA, May 2024.
7. **Jacob M. Chen**, Daniel Malinsky, and Rohit Bhattacharya. "Causal Inference with Outcome Dependent Missingness and Self-Censoring." Poster spotlight talk at Uncertainty in Artificial Intelligence (UAI), Pittsburgh, PA, July 31 - August 4, 2023.
8. **Jacob M. Chen**. "Causal Inference with Outcome Depending Missingness and Self-Censoring." Thesis presentation and defense before the computer science department at Williams College, Williamstown, MA, May 2023.
9. **Jacob M. Chen**, Daniel Malinsky, and Rohit Bhattacharya. "Causal Inference with Outcome Dependent Missingness and Self-Censoring." Poster presentation at the American Causal Inference Conference (ACIC), Austin, TX, May 2023.
10. **Jacob M. Chen** and Rohit Bhattacharya. "On Covariate Adjustment in Missing Not at Random Models." Poster presentation at the American Causal Inference Conference (ACIC), Berkeley, CA, May 2022.

Honors & Awards

- *Finalist for the CRA Undergraduate Research Award*, Computing Research Association (CRA), 2024
- *Sam Goldberg Prize*, Williams College, 2023; Awarded for the best thesis presentation in computer science.
- *Sigma Xi Society*, Williams College, 2023; Inducted into the Sigma Xi Society for excellence in research.
- *Dean's List*, Williams College, 2019-2023

Professional Activities and Services

- *Reviewer*, 42nd Conference on Uncertainty in Artificial Intelligence (UAI), 2026.
- *Reviewer*, Advances in Neural Information Processing Systems 40 (NeurIPS), 2026.
- *Reviewer*, Advances in Neural Information Processing Systems 39 (NeurIPS), 2025. Recognized as Top Reviewer.
- *Reviewer*, npj Digital Medicine, 2025.
- *Reviewer*, 41st Conference on Uncertainty in Artificial Intelligence (UAI), 2025.
- *Reviewer*, 40th Conference on Uncertainty in Artificial Intelligence (UAI), 2024.

Teaching Experience

Residential Teaching Fellow, Phillips Exeter Academy, Exeter, NH — June – Aug. 2022

- Utilized Harkness pedagogy (discussion and project-based learning) to instruct high school students in three courses – Introduction to Computer Science, Mobile App Development, and Game Programming – at a boarding summer school program. I also served as a dorm faculty for boarding high school boys where I provided guidance on navigating the program academically and socially.

Teaching Assistant in Computer Science Department, Williams College — Feb. 2021 – May 2023

- Data Structures & Advanced Programming (Spring '21), Algorithm Design & Analysis (Spring '22), Introduction to Computer Science (Fall '21, Fall '22, & Spring '23)

Lanesborough Elementary School Teaching Fellow, Lanesborough, MA Jan. 2022 – May 2023
- Assisted in a 4th grade classroom with math and science courses and a kindergarten classroom with an English as a second language student. Gave a presentation introducing Taiwan and its culture to elementary school students.

Teacher for AP Computer Science, Nuts Institute, Hsinchu, Taiwan July 2020 – Jan. 2021
- Taught fundamentals of Java to high school students with no previous programming experience in preparation for the AP Computer Science A exam. Prepared my own class materials and lesson plans.

Work Experience

Software Engineer, Seknova Biotechnology Co., Taiwan June – Aug. 2019
- Designed and developed independently Windows graphical user interface tools in Java and developed an Android App for an external blood glucose monitoring device.